

CHEATSHEET · QUICK REFERENCE

Lab 19 — Physical and Environmental Security Walk-Through

Layer physical access controls from perimeter to server hall, compare locks, read environmental sensors, and complete a physical-security audit checklist.

Environmental sensor commands

```
sensors-detect --auto
Detect hardware sensors

sensors
CPU / mainboard temperatures

smartctl -a /dev/sda | grep -i temperature
Drive temperature
```

Defence-in-depth layers

Layer (outside->in)	Control
Property perimeter	Fencing, bollards
Building	Datacenter camouflage, reflective glass
Lobby	Guards, cameras
Inner doors	Card reader + biometric (2-factor)
Between zones	Mantrap / airlock
Server hall	Locked cabinets, safes for KMS/HSM

Locks and replay resistance

Lock type	Authenticator	Replay-resistant?
Mechanical key	Physical key	No
Card reader	Magstripe / prox	No (clonable)
RFID	Badge (+ PIN)	Partly
Biometric	Fingerprint/iris/face	Yes

Environmental controls

Control	Target
Fire suppression	Inert gas (FM-200, Novec 1230)
HVAC	18-27 C, 40-60% RH (ASHRAE)
Sensors	Temp, humidity, smoke, leak, door
Cameras	Retain footage 30-90 days